

# Energy Conversion I

alexis.martin@ensea.fr  
Desk D216

## Layout

- Non isolated choppers
- Switch mode power supplies
- **Power components**
  - Passive components
    - Inductor
    - Transformer
    - Capacitor
  - Active components
    - Diodes
    - Transistors
  - Protections
    - Fuses
    - Thermistors
    - ESD protection



Passive components

Active components

Protections

Passive components



Passive components

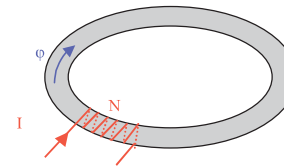
Inductor  
Transformer  
Capacitor

Active components

Protections



Iron core Inductor



$\phi = B \cdot S$  : elementary flux (real flux)

$\Phi = N \cdot \phi = L \cdot I$  : "total flux" (sawn by the electrical circuit)

$$A_L = \frac{1}{\int_{circuit} \frac{dl}{\mu S}} \approx \frac{\mu S}{l_{circuit}}$$

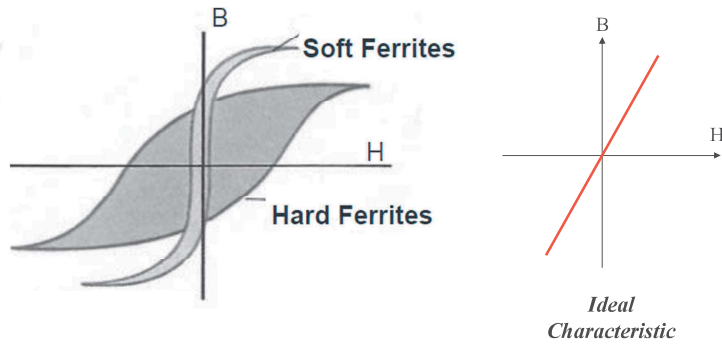
$$\int_{circuit} H \cdot dl = N \cdot I$$

$$H = \frac{B}{\mu}$$

$$N \cdot I = \left[ \int_{circuit} \frac{dl}{\mu \cdot S} \right] \cdot \varphi = R \cdot \varphi$$

$$L_1 = \frac{N^2}{R} = N^2 \cdot A_L$$

## Hysteresis



$$\mu_r = \frac{1}{\mu_0} \frac{\partial B}{\partial H}$$

EMC BASICS, Würth Elektronik

Alexis MARTIN 5

## Losses



Value of relative permeability (and so, inductance) depends on :

- Core material
- Temperature
- Current
- Frequency
- Pressure

$R = R_c + R_I$  : Copper losses (wires) + iron losses (magnetic material)

Alexis MARTIN 6

## Iron core Inductor

- Core material

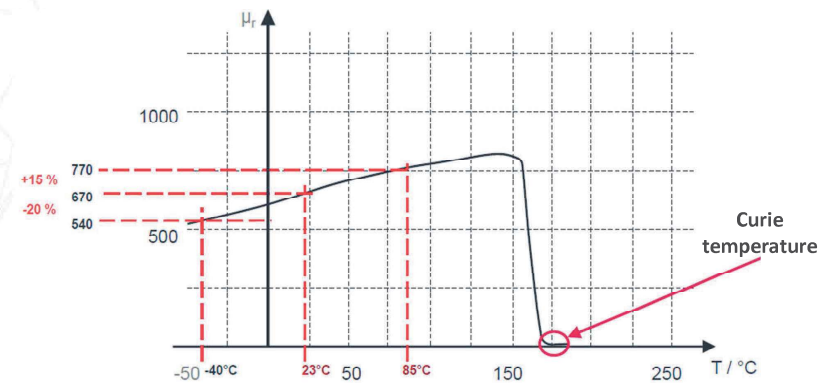
Typical value ranges

- Iron powder (Fe):  $\mu_r = 50 \dots 150$
- Nickel-zinc ferrite ( $\text{Ni}_x\text{Zn}_{1-x}\text{Fe}_2\text{O}_4$ ):  $\mu_r = 50 \dots 1500$
- Manganese-zinc ferrite ( $\text{Mn}_x\text{Zn}_{1-x}\text{Fe}_2\text{O}_4$ ):  $\mu_r = 2000 \dots 10000$
- Nanocrystalline material (Fenc):  $\mu_r = 5000 \dots 95000$

EMC BASICS, Würth Elektronik

## Iron core Inductor

- Temperature



EMC BASICS, Würth Elektronik

Passive components  
Inductor  
Transformer  
Capacitor

Active components

Protections



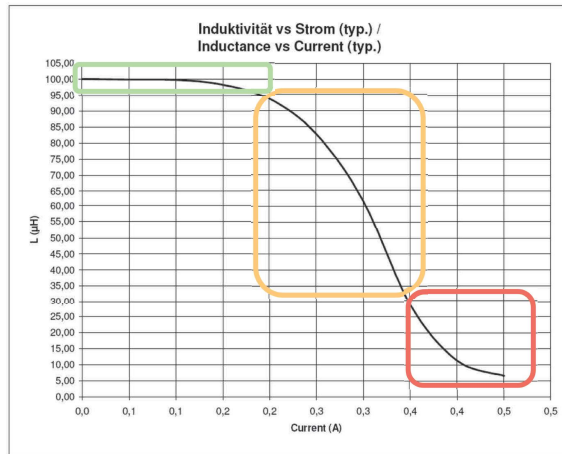
Iron core Inductor

- Current

Safe Operation Area

Critical Operating Area

Total Saturation Area



POWER INDUCTOR SELECTION, Würth Elektronik

Passive components  
Inductor  
Transformer  
Capacitor

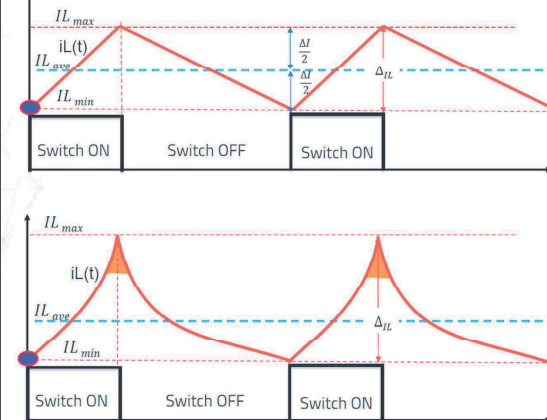
Active components

Protections

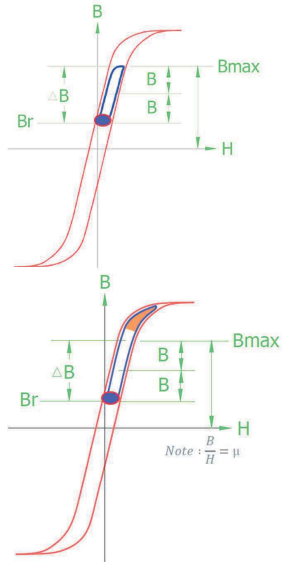


Iron core Inductor

- Current



POWER INDUCTOR SELECTION, Würth Elektronik



Passive components  
Inductor  
Transformer  
Capacitor

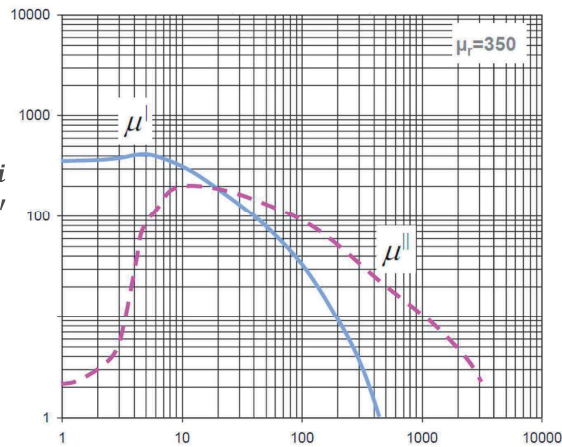
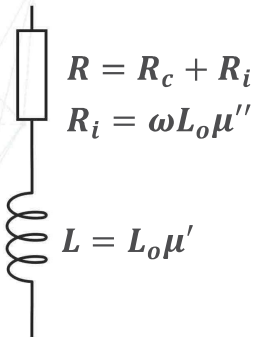
Active components

Protections



Iron core Inductor

- Frequency



EMC BASICS, Würth Elektronik

Passive components  
Inductor  
Transformer  
Capacitor

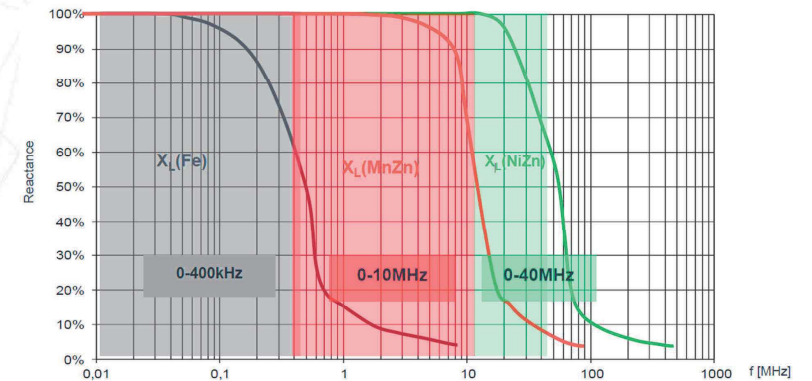
Active components

Protections



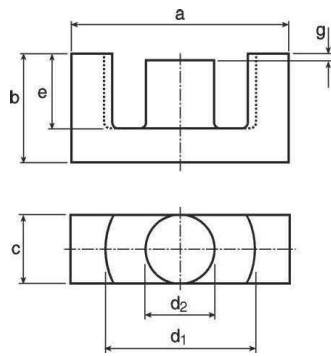
Iron core Inductor

- Frequency



EMC BASICS, Würth Elektronik

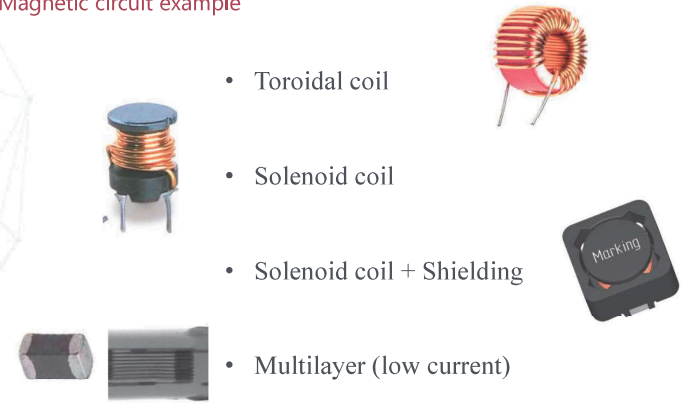
### Magnetic circuit example



ETD series



### Magnetic circuit example



- Toroidal coil
- Solenoid coil
- Solenoid coil + Shielding
- Multilayer (low current)

### Stored energy – Air-gap

Magnetic energy density stored

$$\omega_{mag} = \frac{B^2}{2 \cdot \mu_0 \cdot \mu_r}$$

Example : Ferrinox B52

$B = 0,4 \text{ T}$

$\mu = \mu_0 \cdot \mu_r = 2,5 \cdot 10^{-3} \text{ H.m}^{-1}$

$$\omega_{mag} = 32 \text{ J/m}^3$$

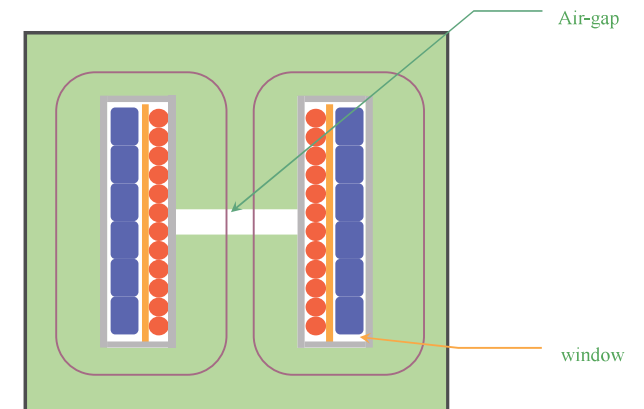
Air-gap :  $\mu_0 = 4\pi \cdot 10^{-7} \text{ H.m}^{-1}$

$$\omega_{mag} = 65 \text{ kJ/m}^3$$



Store energy in the air-gap

### With air-gap



## Passive components

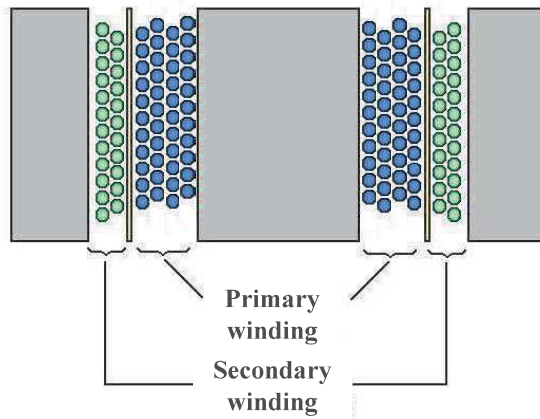
Inductor  
Transformer  
Capacitor

## Active components

## Protections



Closed core



Alexis MARTIN 17

## Passive components

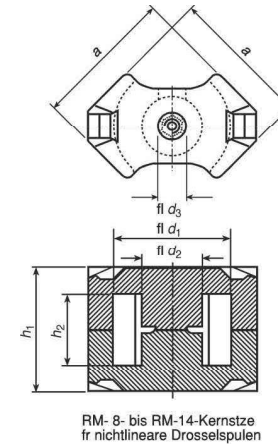
Inductor  
Transformer  
Capacitor

## Active components

## Protections



Magnetic pot core



*RM series*

Alexis MARTIN 18

## Passive components

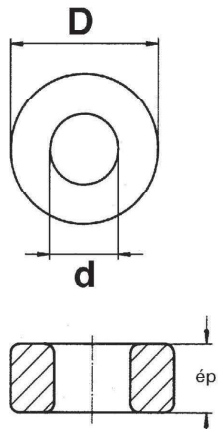
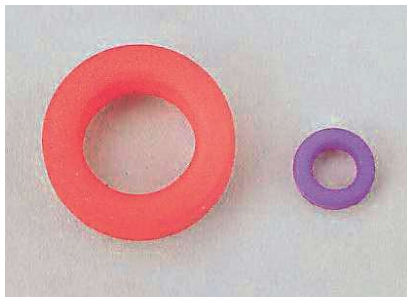
Inductor  
Transformer  
Capacitor

## Active components

## Protections



Torus



Alexis MARTIN 19

## Passive components

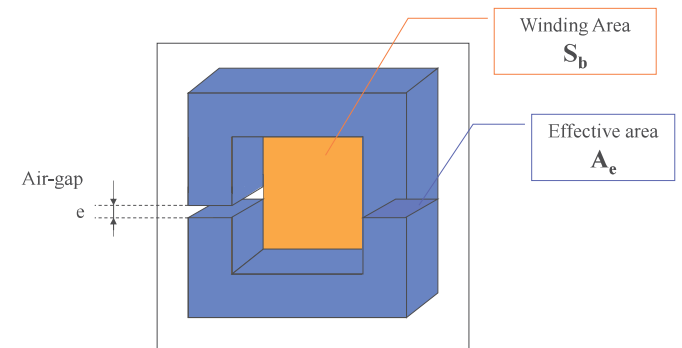
Inductor  
Transformer  
Capacitor

## Active components

## Protections



Magnetic circuits parameters



Inductance factor:  $\mathbf{A_L}$  = Inductance obtained with ONE loop

Alexis MARTIN 20

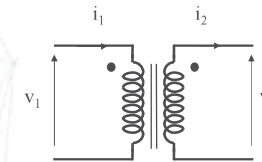
## Critical constraints

Induction in the material:  $B < B_{sat}$   
0,2 to 0,5 T for classical materials

Current density in the winding:  $\delta < \delta_{max}$   
5 A/mm<sup>2</sup> for copper

**Sizing : product  $A_e S_b$**

## Transformer



Primary :  $N_1$  turns  
Secondary :  $N_2$  turns

Winding direction

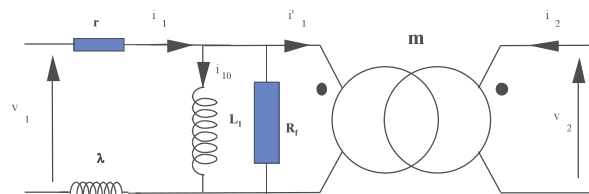
$$v_1 = N_1 \cdot \frac{d\phi}{dt} \quad v_2 = N_2 \cdot \frac{d\phi}{dt} \quad m = \frac{N_2}{N_1} = \frac{v_2}{v_1}$$

$$\oint H \cdot dl = R \cdot \phi = N_1 \cdot i_1 - N_2 \cdot i_2$$

Magnetizing current  $i_{10}$  : Hopkinson's law  $R \cdot \phi = N_1 \cdot i_{10}$

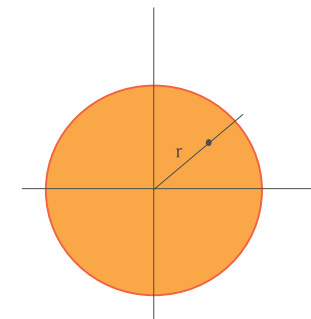
$$\Phi = L_1 \cdot i_{10} \quad L_1 = \frac{N_1^2}{R} \quad i_1 - m \cdot i_2 = i_{10} \quad v_1 = L_1 \cdot \frac{di_{10}}{dt}$$

## Transformer : equivalent scheme



$L_1$  : Magnetizing inductance (at primary side)  
 $R_f$  : ferromagnetic losses  
 $r$  : Joule loss (or ohmic loss)  
 $\lambda$  : leakage inductance

## Skin effect

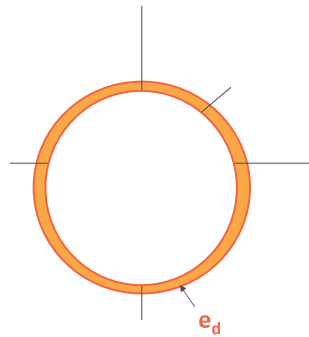


$$j(r) = j_0 \cdot e^{-i\pi \frac{r}{\delta}}$$

$$e_d = \frac{1}{\sqrt{\pi \cdot \sigma \cdot \mu \cdot f}}$$

- $e_d$  : skin depth thickness
- $\sigma$  : conductivity
- $\mu$  : permeability
- $f$  : frequency

Skin effect



Copper at 25°C :  $\sigma = 6 \cdot 10^7 \text{ S/m}$

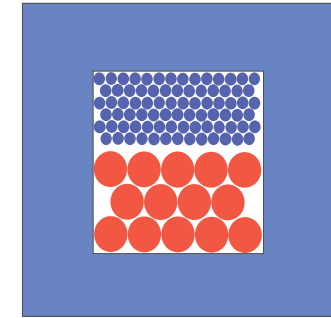
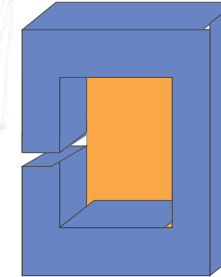
$$e_d \approx \frac{66}{\sqrt{f}}$$

- 50 Hz :  $e_d = 9,4 \text{ mm}$
- 100 kHz :  $e_d = 0,21 \text{ mm}$

=> *Multi strand wire (Litz wire)*

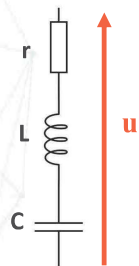
For AC part of current !

Winding



$$k_B = 0,7$$

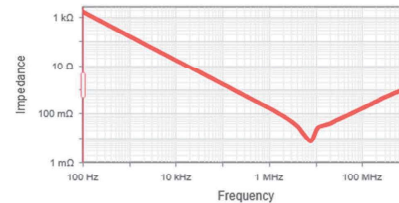
Capacitors



*Losses in the capacitor*

$$P = r \cdot I_{\text{eff}}^2$$

Rated voltage:  $U_{\text{eff}}$



$r$  : ESR (Equivalent Series Resistor)  
 $L$  : ESL (Equivalent Series Inductor)

*Multiple technologies : depends on the frequency*

Capacitors

| Polarized                       |                | Unpolarized                          |              |
|---------------------------------|----------------|--------------------------------------|--------------|
| Aluminum                        | Solid Tantalum | MLCC : Multilayer ceramic capacitors | Film         |
|                                 |                |                                      |              |
| High value                      |                | High voltage                         |              |
| High capacitance                |                | Small size                           | Self healing |
| Lifetime depends on temperature | More expensive | Low ESR & ESL                        |              |
|                                 | Small size     | "lower" capacitance                  |              |

Capacitor Selection Guidelines for Analog Devices  
Fundamentals of Capacitor Technology and their Applications (Würth Elektronik)  
Capacitor Selection Guide (Murata)

## Capacitors

### MLCC :

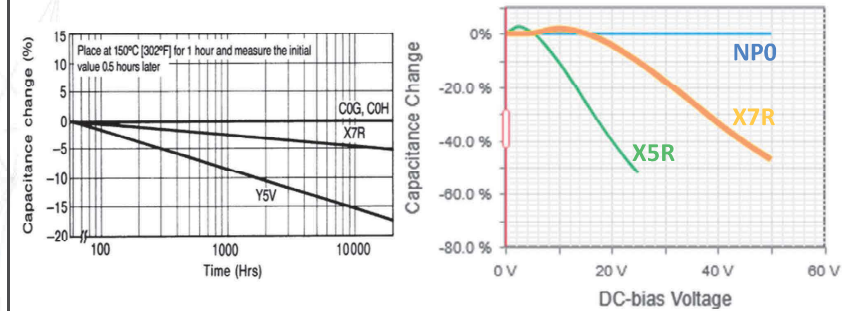
- Class I ceramic capacitor (NP0, C0G ...)
  - Made from ceramic material **not sensitive to temperature** (TiO<sub>2</sub> ...)
  - Low permittivity -> **low capacitance value**
  - Next to no aging
- Class II ceramic capacitor (X5R, X7R, Y5V ...)
  - Made from ceramic materials that are derived from a barium titanate (BaTiO<sub>3</sub>) base : **sensitive to temperature**
  - High permittivity : **high capacitance value**
  - High voltage dependency

MLCC Application Notes (TDK)  
Fundamentals of Capacitor Technology and their Applications (Würth Elektronik)

Alexis MARTIN 29

## Capacitors

### MLCC :

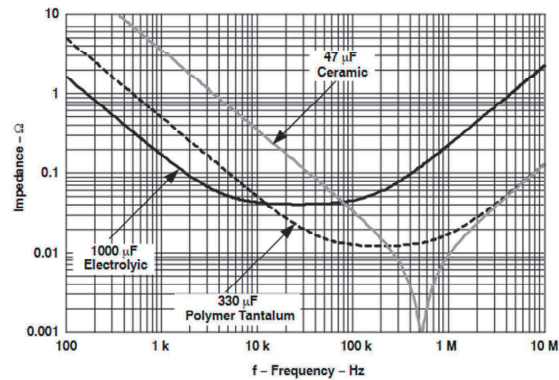


MLCC Application Notes (TDK)  
Fundamentals of Capacitor Technology and their Applications (Würth Elektronik)

Alexis MARTIN 30

## Capacitors

### ESR and ESL variations

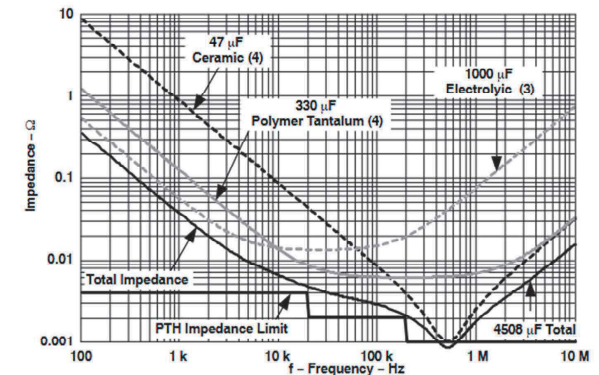


Input and Output Capacitor Selection (Texas Instrument)

Alexis MARTIN 31

## Capacitors

### Multiple technologies in parallel



Input and Output Capacitor Selection (Texas Instrument)

Alexis MARTIN 32

Passive components

Active components

Protections



Active components

Passive components

Active components

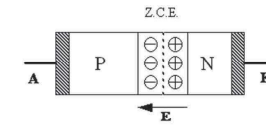
Diodes  
MOS transistor  
IGBT transistor

Protections

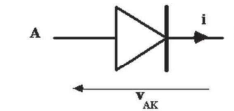


Active components

PN junction



Diode



$$i = I_S \left( e^{\frac{q \cdot v_{AK}}{k \cdot T}} - 1 \right)$$

Passive components

Active components

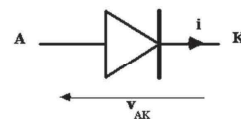
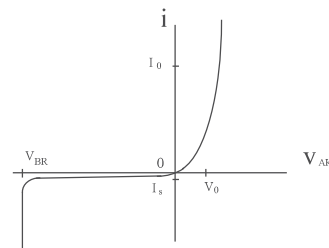
Diodes  
MOS transistor  
IGBT transistor

Protections



Diode : static characteristic

$I_0$  : Average forward current, or  $I_P$ ,  $I_{AV}$   
 $I_{FM}$  : Peak forward current  
 $I_{FSM}$  : Peak forward surge current  
 $I_{RRM}$  : Reverse repetitive current



$V_0$  : Forward voltage or  $V_{FM}$   
 $V_{BR}$  : Breakdown reverse voltage  
 $V_{RRM}$  : Peak repetitive reverse voltage

Passive components

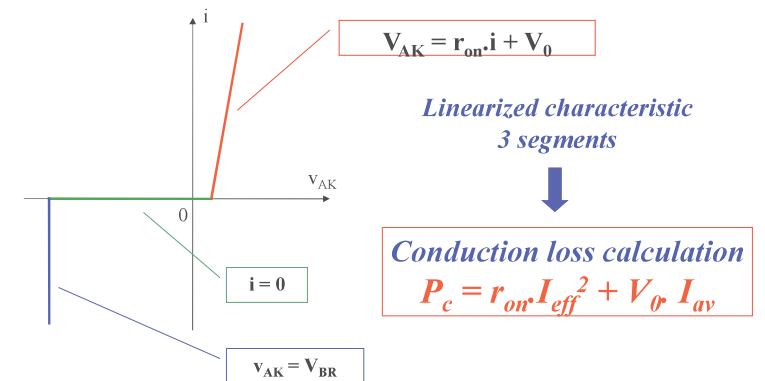
Active components

Diodes  
MOS transistor  
IGBT transistor

Protections



Diode : simplified static characteristics



## Passive components

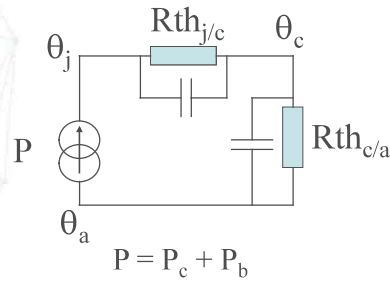
## Active components

Diodes  
MOS transistor  
IGBT transistor

## Protections

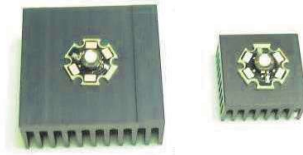


Heat sink



$$\theta_j < \theta_{jmax}$$

$Rth_{c/a}$  calculation



Alexis MARTIN 37

## Passive components

## Active components

Diodes  
MOS transistor  
IGBT transistor

## Protections



Specific diode

- Low threshold diode: Schottky diode
- High voltage diode
- Power Zener diodes (« Transient-voltage-suppression diode »)
- Fast diodes (low  $Q_R$ )
- Light-emitting diodes (high threshold)

Alexis MARTIN 38

## Passive components

## Active components

Diodes  
MOS transistor  
IGBT transistor

## Protections

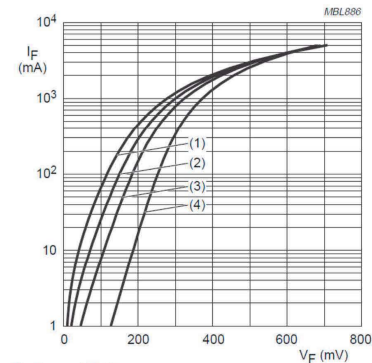


Schottky diode



Low  $V_F$  -> Low conduction losses

- Used as diodes in converters (instead of PN diodes)
- "low" voltage (max~100V)



- (1)  $T_{amb} = 125\text{ °C}$ .
- (2)  $T_{amb} = 100\text{ °C}$ .
- (3)  $T_{amb} = 75\text{ °C}$ .
- (4)  $T_{amb} = 25\text{ °C}$ .

Fig.2 Forward current as a function of forward voltage; typical values.

Alexis MARTIN 39

## Passive components

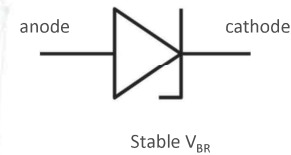
## Active components

Diodes  
MOS transistor  
IGBT transistor

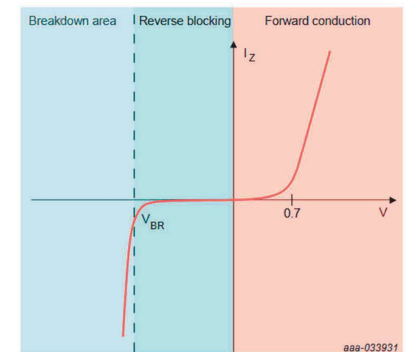
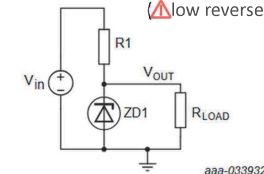
## Protections



Zener diode



- Low-cost stabilized voltage
- Signal overvoltage clamping (Δ low reverse current)



I-V characteristic of a Zener diode

| BZX884S | Sel | Working voltage $V_Z$ (V) |       | Differential resistance $r_{dif}$ (Ω) |     |                     |     |
|---------|-----|---------------------------|-------|---------------------------------------|-----|---------------------|-----|
|         |     | $I_Z = 5\text{ mA}$       |       | $I_Z = 1\text{ mA}$                   |     | $I_Z = 5\text{ mA}$ |     |
| 18      | B   | Min                       | Max   | Typ                                   | Max | Typ                 | Max |
|         |     | 17.60                     | 18.40 | 50                                    | 225 | 10                  | 45  |
|         | C   | 16.80                     | 19.10 |                                       |     |                     |     |

AN90031 (Nexperia)

Alexis MARTIN 40

## TVS diode

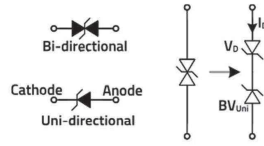
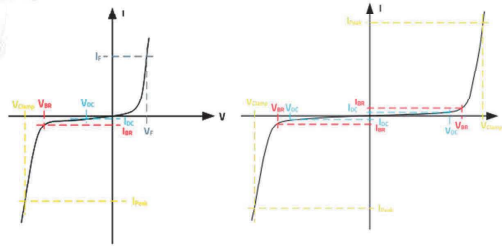


Figure 1: Electrical symbols for the uni- and bidirectional TVS diodes.  
Right figure: Functional equivalent of the bidirectional TVS diode.



ANP143 (Würth Elektronik)

## TVS : Transient Voltage Suppressors

- Protection against high voltage spikes
- ESD protection (USB ...)
- Overvoltage clamping (high peak pulse current)

### WE-TVSP Power TVS Diode, 824500101

| Properties                     | Test conditions                   | Value          | Unit    | Tol. |
|--------------------------------|-----------------------------------|----------------|---------|------|
| DC Operating Voltage           | $V_{DO}$                          | 10             | V       | max. |
| (Reverse) Breakdown Voltage    | $V_{BR}$ 1 mA                     | 11.7           | V       | ±5%  |
| Clamping Voltage               | $V_{CL}$ $I_{CL}$                 | 17             | V       | max. |
| (Reverse) Peak Pulse Current   | $I_{RR}$ 10/1000 $\mu$ s          | 23.5           | A       | max. |
| (Forward) Peak Pulse Current   | $I_{FR}$ 10/1000 $\mu$ s          | 60             | A       | max. |
| Leakage Current                | $I_{L}$ 1V                        | 5              | $\mu$ A | max. |
| Steady State Power Dissipation | $P_{SS}$ $T_J = 50^\circ\text{C}$ | 3.3            | W       | max. |
| Power Dissipation              | $P_{AV}$ 10/1000 $\mu$ s          | 400            | W       | max. |
| Polarity                       |                                   | Unidirectional |         |      |

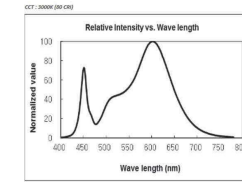
Alexis MARTIN 41

## LED diode

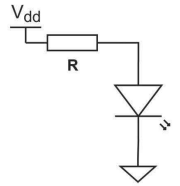


LM301B (Samsung)

b) Electro-optical Characteristics ( $I_F = 65 \text{ mA}$ ,  $T_s = 25^\circ\text{C}$ )



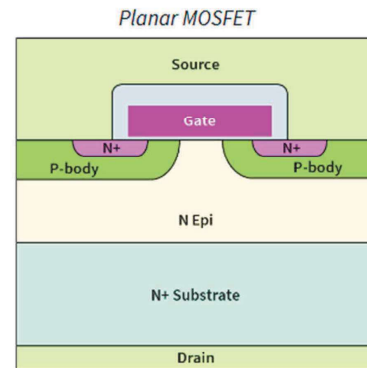
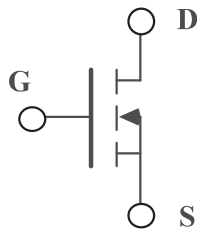
« White » LED



| Item                                          | Unit               | Rank | Bin | Min. | Typ. | Max. |
|-----------------------------------------------|--------------------|------|-----|------|------|------|
| Forward Voltage ( $V_F$ )                     | V                  | XA   | AY  | 2.6  | -    | 2.7  |
|                                               |                    |      | AZ  | 2.7  | -    | 2.8  |
|                                               |                    |      | A1  | 2.8  | -    | 2.9  |
| Reverse Voltage ( $V_R$ )                     | V                  |      |     | 0.7  | -    | 1.2  |
| Color Rendering Index ( $R_a$ )               | -                  |      |     | 80   | -    | -    |
| Thermal Resistance (junction to solder point) | $^\circ\text{C/W}$ |      |     | -    | 7.5  | -    |
| Beam Angle                                    | $^\circ$           |      |     | -    | 120  | -    |

Alexis MARTIN 42

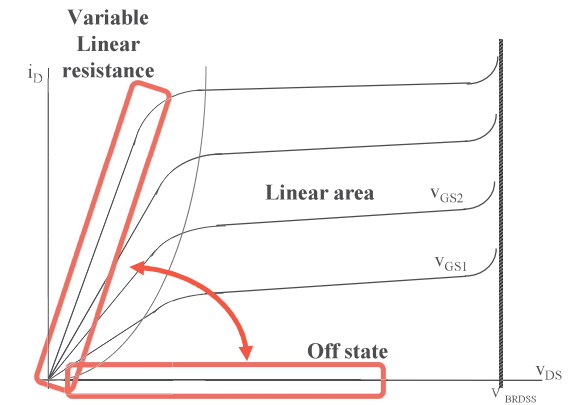
## Power MOSFET



Designing with power MOSFETs (Infineon)

Alexis MARTIN 43

## MOSFET characteristics



Alexis MARTIN 44

Passive components

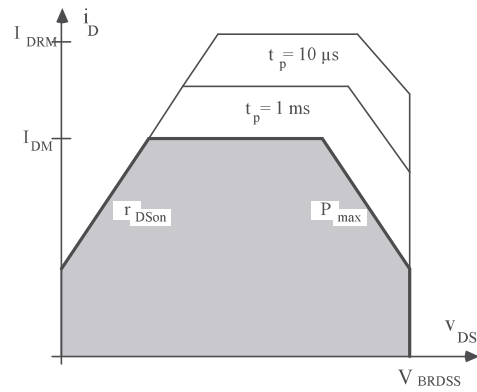
Active components

Diodes  
MOS transistor  
IGBT transistor

Protections



MOS safety area



Alexis MARTIN 45

Passive components

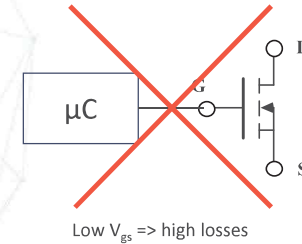
Active components

Diodes  
MOS transistor  
IGBT transistor

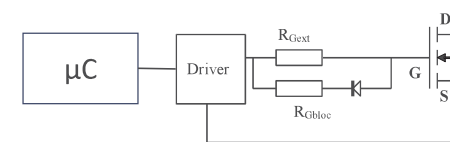
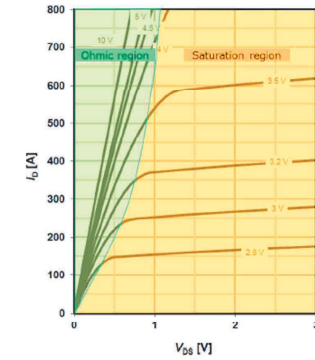
Protections



Control specificity



Low  $V_{gs} \Rightarrow$  high losses



Alexis MARTIN 46

Passive components

Active components

Diodes  
MOS transistor  
IGBT transistor

Protections



MOSFET Gate Driver LTC7065 (Analog Devices)

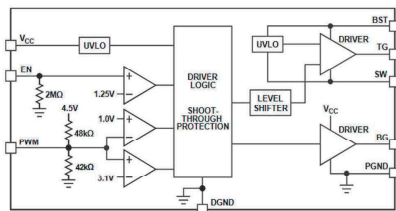
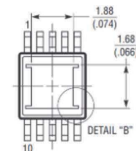


Figure 15. Block Diagram

10-LEAD MSOP Package



TYPICAL APPLICATION

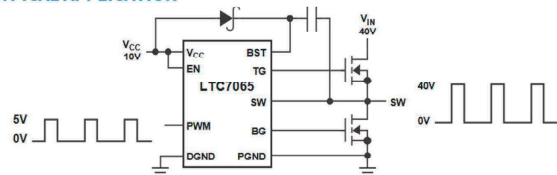


Figure 1. 40V Half-Bridge Driver

Alexis MARTIN 47

Passive components

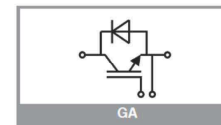
Active components

Diodes  
MOS transistor  
IGBT transistor

Protections



Power IGBT



| Absolute Maximum Ratings |                           |        |      |
|--------------------------|---------------------------|--------|------|
| Symbol                   | Conditions                | Values | Unit |
| IGBT                     |                           |        |      |
| $V_{CES}$                | $T_J = 25^\circ\text{C}$  | 1200   | V    |
| $I_C$                    | $T_J = 175^\circ\text{C}$ | 1305   | A    |
|                          | $T_c = 80^\circ\text{C}$  | 1003   | A    |
| $I_{Cnom}$               |                           | 900    | A    |

Alexis MARTIN 48

Passive  
components

Active  
components

Diodes  
MOS transistor  
IGBT transistor

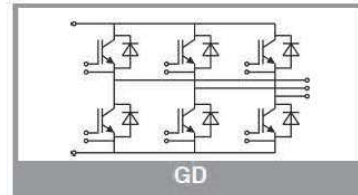
Protections



Modules



1200 V – 39 A at 25°C –  $t_f < 200\text{ns}$



1200 V – 231 A at 25°C –  $t_f < 500\text{ns}$

Alexis MARTIN 49

Passive  
components

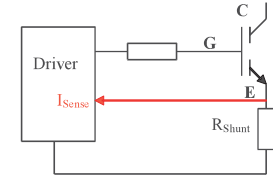
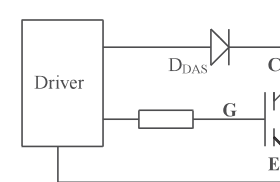
Active  
components

Diodes  
MOS transistor  
IGBT transistor

Protections



Control specificity



Alexis MARTIN 50

Passive  
components

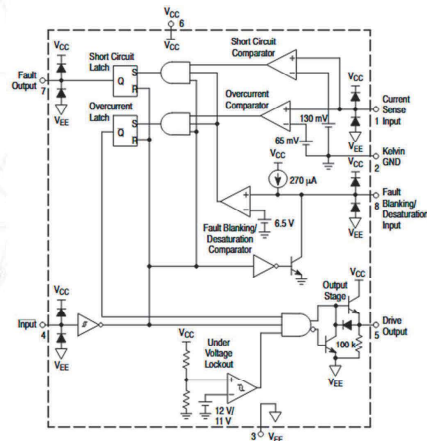
Active  
components

Diodes  
MOS transistor  
IGBT transistor

Protections



IGBT Gate Driver MC33153 (Onsemi)



This device contains 133 active transistors.  
Figure 1. Representative Block Diagram

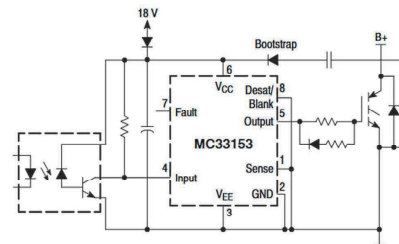
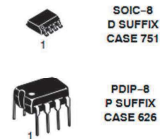


Figure 35. Basic Application



Alexis MARTIN 51

Passive  
components

Active  
components

Protections



Protections

Passive components

Active components

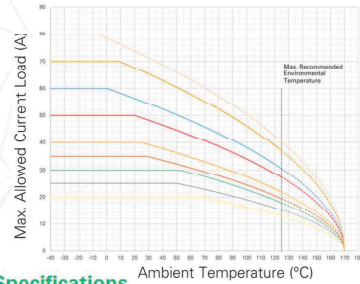
Protections

Fuses  
Thermistors  
ESD protection



Fuses

- Overcurrent protection



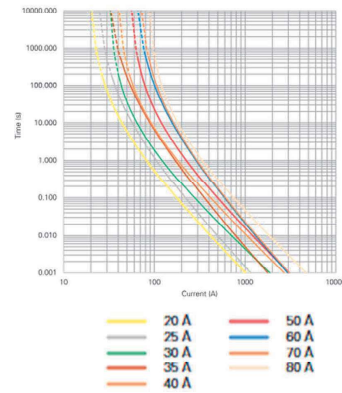
Specifications

|                      |                  |
|----------------------|------------------|
| Voltage Rating:      | 32 V DC          |
| Interrupting Rating: | 1000 A @ 32 V DC |

| Part Number | Current Rating (A) | Housing Material Color | Test Cable Size (mm <sup>2</sup> ) | Typ. Voltage Drop (mV) | Typ. Cold Resistance (mΩ) | Typ. I <sup>2</sup> t (A <sup>2</sup> s) |
|-------------|--------------------|------------------------|------------------------------------|------------------------|---------------------------|------------------------------------------|
| 0299020_    | 20                 | Yellow                 | 4                                  | 70                     | 3.1                       | 1100                                     |

MAXI Series Blade Fuses – Rated 32V (Littlefuse)

Time-Current Characteristic



Passive components

Active components

Protections

Fuses  
Thermistors  
ESD protection



Fuses

Depending on the reaction time needed, five classes are available on the market (IEC 60127-1 standard):

- FF: very quick acting
- F: quick acting**
- M: medium time-lag
- T: time-lag / time delay**
- TT: long time-lag

| Fast acting                                                                                                                                                                    | Time lag / time delay                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>No intentional time-delay feature</li> <li>Fast acting</li> <li>Hundreds of milliseconds acting</li> <li>Non inductive loads</li> </ul> | <ul style="list-style-type: none"> <li>Few seconds delay (motor start ...)</li> <li>Motor and general protection</li> </ul> |

How to Select the Right Fuse (Vishay) + Fuse Fundamentals (Littlefuse)

Passive components

Active components

Protections

Fuses  
Thermistors  
ESD protection



NTC thermistors

NTC : Negative Temperature Coefficient

Commonly referred to as Inrush Current Limiters (ICLs)

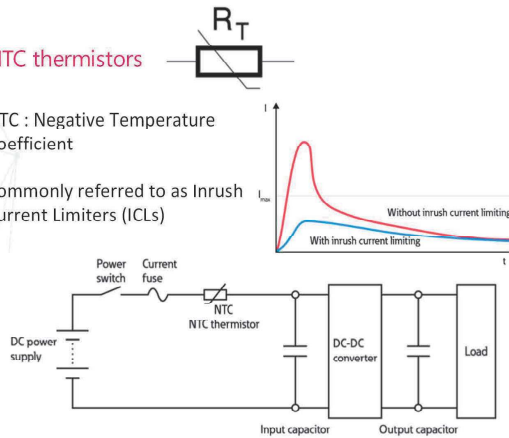
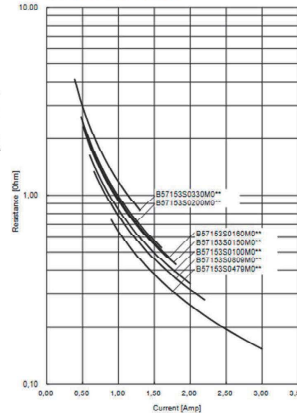


Figure 4: Sample application circuit with power NTC thermistor for a DC-DC converter.

Technical Note ELX1247 (Eaton) + NTC Inrush Current Limiters Application notes (TDK)

Resistance versus current



Alexis MARTIN 55

Passive components

Active components

Protections

Fuses  
Thermistors  
ESD protection



PTC thermistors

PTC : Positive Temperature Coefficient

- Overcurrent protection
- Temperature sensing

➤ High voltage

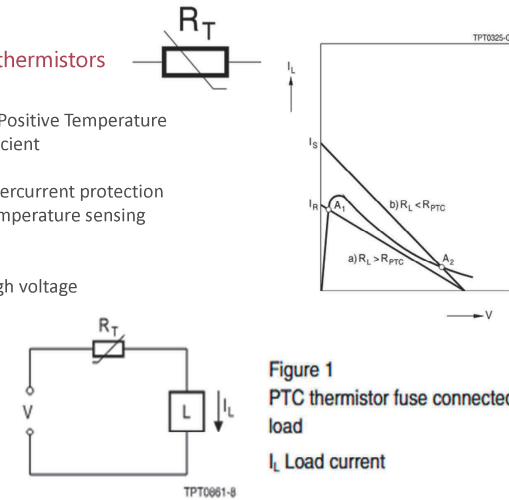


Figure 1  
PTC thermistor fuse connected in series with the load  
I<sub>L</sub> Load current

PTC Thermistors Application notes (TDK)

Figure 2  
Operating states of a PTC thermistor for overcurrent protection  
a) Rated operation  
b) Overload operation  
R<sub>L</sub> Load resistance  
I<sub>S</sub> Switching current  
I<sub>N</sub> Rated current  
R<sub>PTC</sub> PTC resistance

Alexis MARTIN 56

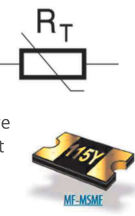
Passive components

Active components

Protections  
Fuses  
Thermistors  
ESD protection



PPTC thermistors



PPTC : Polymeric Positive Temperature Coefficient

Also known as **polyfuse**

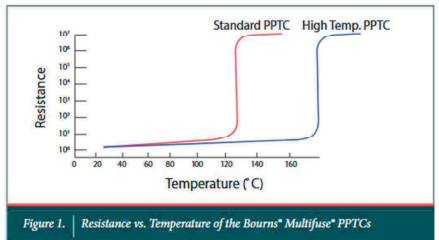
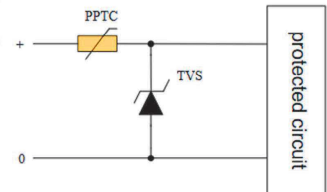


Figure 1. Resistance vs. Temperature of the Bourns® Multifuse® PPTCs

| Part Number | Marking | I <sub>hold</sub> (A) | I <sub>trip</sub> (A) |
|-------------|---------|-----------------------|-----------------------|
| 0402L010SL  | -       | 0.10                  | 0.30                  |

Application notes K1467 (Bourns) + POLY-FUSE® and PolySwitch® SMD PPTC Application (Littlefuse)  
+ Applications of PPTC in Circuit Protections (Yageo)

Passive components

Active components

Protections  
Fuses  
Thermistors  
ESD protection



ESD protection

USB TYPE-C Port

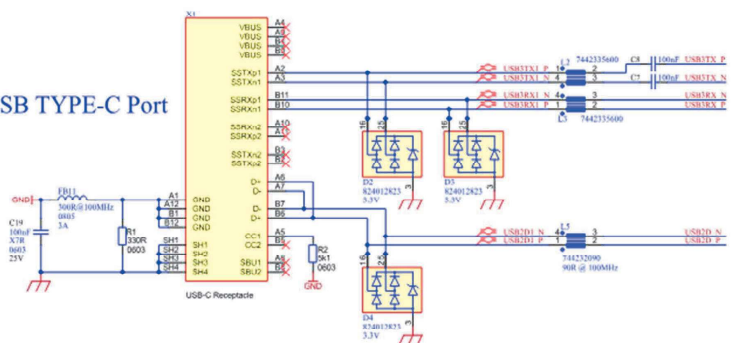


Figure 5: Transient protection at a USB interface with a TVS diode array.